Chapter 12 - Iterative Methods

- To solve simultaneous linear equations.

Linear Systems: Gauss-Siedel

$Ax = b$, Let the system be 3×3.

- If the diagonal elements of A are non-zero:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

1st eq^n $\Rightarrow$ solve for x_1: $$x_1^j = \frac{b_1 - a_{12}x_2^{j-1} - a_{13}x_3^{j-1}}{a_{11}} \quad (1)$$

2nd eq^n $\Rightarrow$ solve for x_2: $$x_2^j = \frac{b_2 - a_{21}x_1^j - a_{23}x_3^{j-1}}{a_{22}} \quad (2)$$

3rd eq^n $\Rightarrow$ " x_3: $$x_3^j = \frac{b_3 - a_{31}x_1^j - a_{32}x_2^j}{a_{33}} \quad (3)$$

$j, j-1$: current and past iterations

initial guess : assume they're all zero.

visit eq^n (1) - (2) - (3) until convergence.

Convergence criterion: $$\varepsilon_{a,i} = \left| \frac{x_i^j - x_i^{j-1}}{x_i^j} \right| \times 100\% \leq \varepsilon_s$$

EX Use Gauss-Siedel iterations to solve:

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$
$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$
$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

(Note: solution is $x_1 = 3$, $x_2 = -2.5$, $x_3 = 7$)

$$x_1 = \frac{7.85 + 0.1x_2 + 0.2x_3}{3} \quad (1)$$

$$x_2 = \frac{-19.3 - 0.1x_1 + 0.3x_3}{7} \quad (2)$$

$$x_3 = \frac{71.4 - 0.3x_1 + 0.2x_2}{10} \quad (3)$$

Start with $x_1 = x_2 = x_3 = 0 \Rightarrow x_1 = 2.61$ from (1)

$x_2 = -2.79$ from (2)

$x_3 = 7.005$ from (3)

2nd iteration: $x_1 = 2.99$

$x_2 = -2.49$

$x_3 = 7.00$

$$\varepsilon_{a,1} = \left| \frac{2.99 - 2.61}{2.99} \right| .100 = 12.5\% \quad (\text{for } x_1)$$

$\varepsilon_{a,2} = 11.8\%$ (for x_2) $\qquad \varepsilon_{a,3} = 0.076$ (for x_3)

Jacobi iteration:

→ Rather than using latest values, use the old values of x's.

→ As new values are generated they are not immediately used but rather retained for the next iteration.

$$x_1^j = \frac{b_1 - a_{12} x_2^{j-1} - a_{13} x_3^{j-1}}{a_{11}} \quad (1)$$

$$x_2^j = \frac{b_2 - a_{21} x_1^{j-1} - a_{23} x_3^{j-1}}{a_{22}} \quad (2)$$

$$x_3^j = \frac{b_3 - a_{31} x_1^{j-1} - a_{32} x_2^{j-1}}{a_{33}} \quad (3)$$

Convergence of Gauss-Siedel

It can be shown that if the following condition holds Gauss-Siedel will converge:

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}| \Rightarrow \text{diagonally dominant systems.}$$

- Sufficient but not necessary condition.
- Convergence is guaranteed if condition is met.
- May sometimes work even if condition is not true.

Nonlinear Systems

$$\left.\begin{aligned} x_1^2 + x_1 x_2 &= 10 \\ x_2 + 3x_1 x_2^2 &= 57 \end{aligned}\right\} \text{ Solve simultaneously}$$

Similar to roots of nonlinear eq^{n}s:

$$\left.\begin{aligned} f_1(x_1, x_2, \dots x_n) &= 0 \\ &\vdots \\ f_m(x_1, x_2, \dots x_n) &= 0 \end{aligned}\right\} \text{ solution: } x\text{'s that make the equations zero.}$$

Successive substitions method. (~ Fixed pt. iteration)

- Each of the nonlinear equations can be solved for one of the unknowns.
- Implement iteratively.

Ex $\left.\begin{aligned} x_1^2 + x_1 x_2 &= 10 \quad (1) \\ x_2 + 3x_1 x_2^2 &= 57 \quad (2) \end{aligned}\right\}$ Solution $x_1 = 2$ and $x_2 = 3$

Solve (1) for $x_1 = \sqrt{10 - x_1 x_2}$

Solve (2) for $x_2 = \sqrt{\dfrac{57 - x_2}{3x_1}}$

iteration 1: $\begin{cases} x_1 = 2.17 \\ x_2 = 2.86 \end{cases}$ iteration 2: $\left.\begin{cases} x_1 = 1.94 \\ x_2 = 3.04 \end{cases}\right\}$ converging